Chapter 1 DNA and genetics

EXTRA WORKSHEET 1: Karyotypes

It is possible to isolate some cells from a developing fetus and photograph the chromosomes in the cells. Scientists use the photograph to create a karyotype and from this they can identify genetic abnormalities.

To create a karyotype, the photograph is cut up and individual chromosomes are arranged into homologous pairs according to their size, as shown in Figure 1. Homologous pairs are chromosomes with the same genes for particular characteristics at the same location along the length of each chromosome. The pairs are numbered for ease of identification. The karyotype that is produced can provide information about the developing foetus.

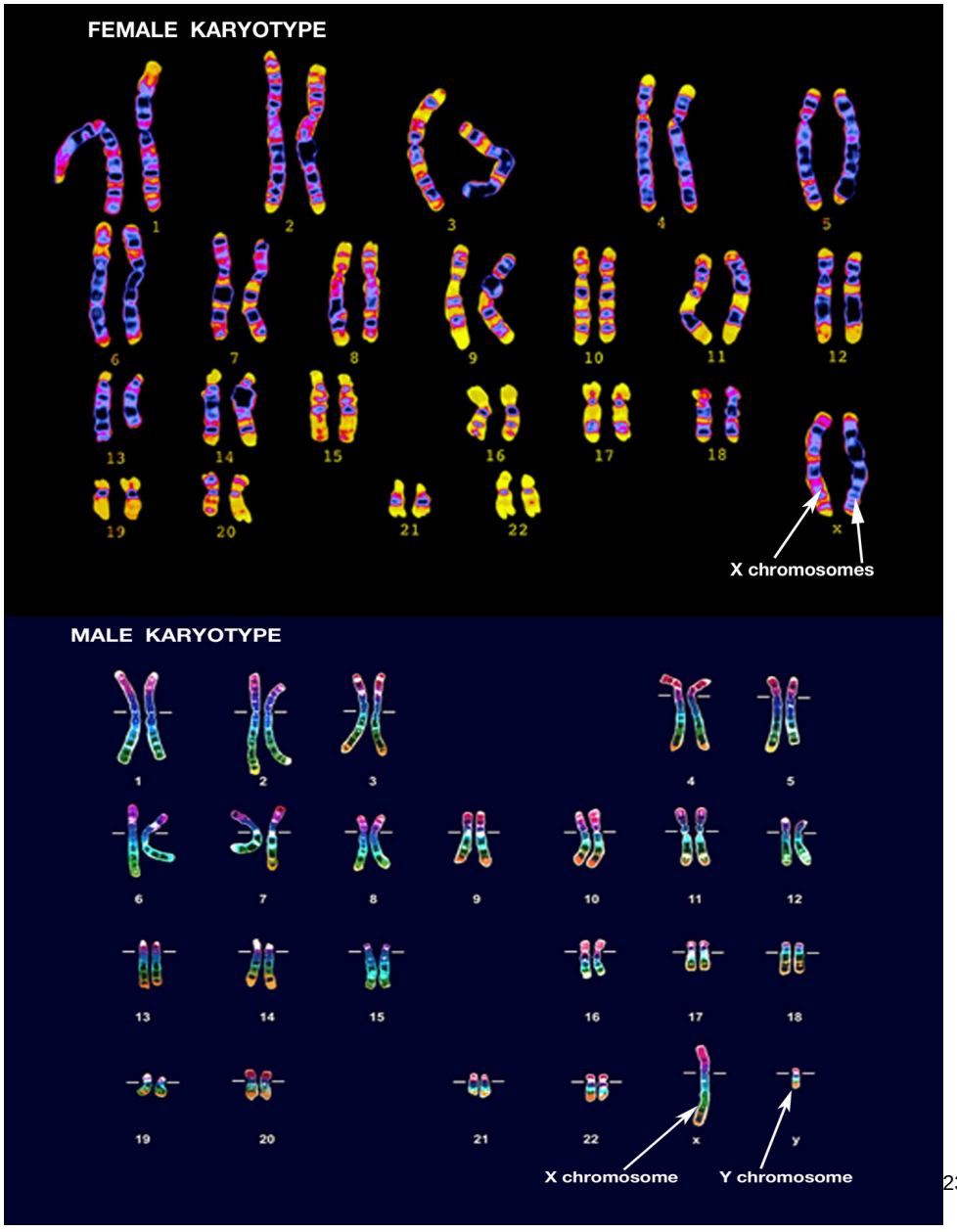
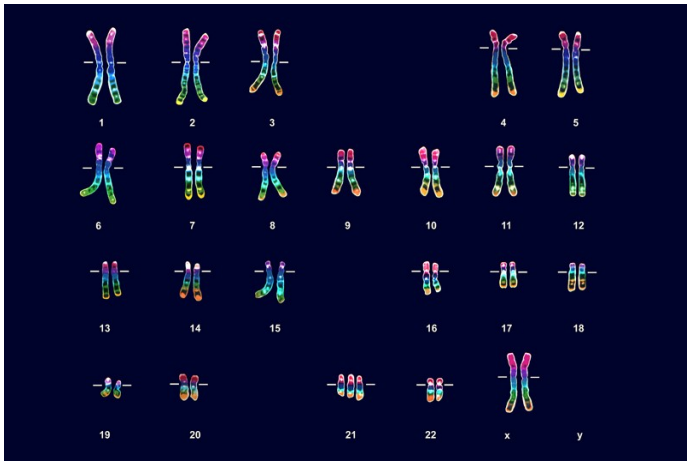
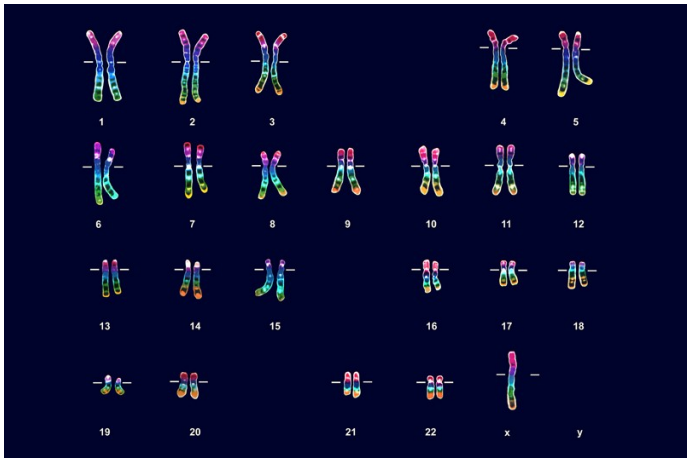
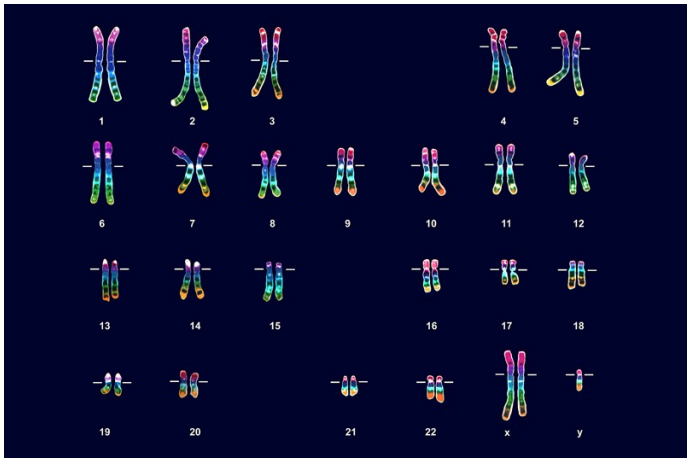


Figure 1
These karyotypes are of a normal male and a normal female. Look for the difference in the sex chromosomes. The Y chromosome is much smaller than the X chromosome.

Questions

1 **Identify** the karyotype that matches the description of the following three syndromes.

Karyotype	Description of Syndrome
	Klinefelter's syndrome: Male with an extra X chromosome (genotype XXY).
	Down syndrome: Extra copy of chromosome 21.
	Turner's syndrome: Absence of a Y or second X chromosome (genotype XO).

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2 Create your own karyotype by cutting up the mixed up picture of chromosomes and then matching the homologous pairs of chromosomes. Glue them in the space below, and then answer the questions.

a **State** the number of homologous pairs of chromosomes in Figure 2.

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b **Deduce** whether this set of chromosomes came from a male or a female.

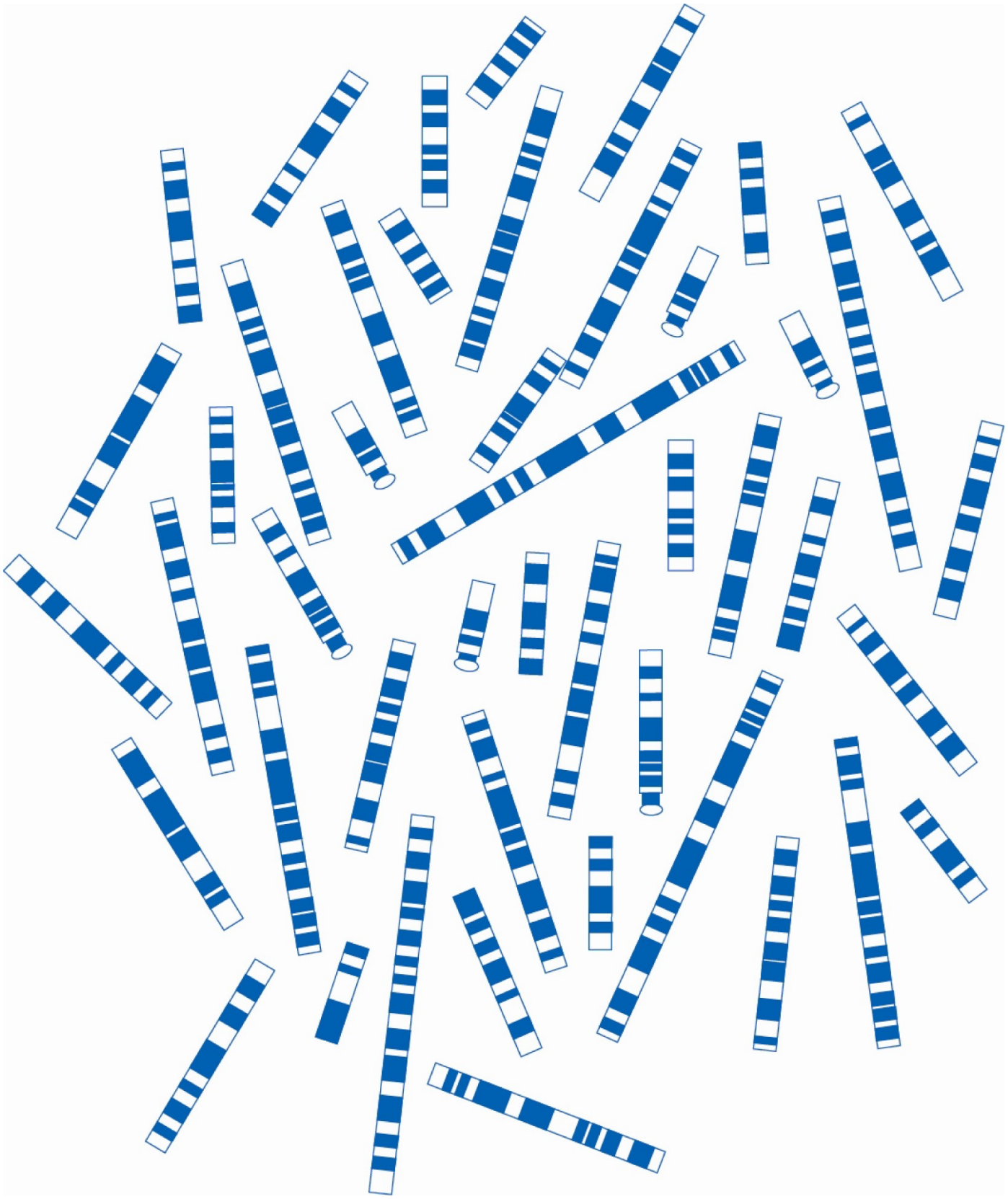
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c **Justify** your response.

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